

CareerPilot AI - Product Requirement Document (PRD)

1. Product Overview

Product Name

CareerPilot AI

Product Vision

CareerPilot AI is an AI-powered career management platform that helps job seekers manage their entire job search journey from resume optimization to application tracking and interview preparation.

The platform combines resume analysis, ATS optimization, job description matching, application tracking, and AI-powered career insights into a single unified dashboard.

Product Mission

Help users increase their chances of getting interviews and job offers by providing actionable AI-driven recommendations throughout the job search process.

2. Problem Statement

Job seekers face multiple challenges during the hiring process:

- Resumes are often rejected by Applicant Tracking Systems (ATS) before reaching recruiters.
- Users do not know which keywords or skills are missing from their resumes.
- Tailoring resumes for every job application is time-consuming.
- Tracking dozens or hundreds of applications becomes difficult.
- Users struggle to identify skill gaps for target roles.
- Interview preparation is often generic and not personalized.

Most existing tools solve only one part of the problem.

CareerPilot AI aims to become a centralized platform that manages the complete job search lifecycle.

3. Goals

Primary Goals

- Help users improve ATS compatibility.

- Help users understand resume strengths and weaknesses.
- Help users compare resumes against job descriptions.
- Help users identify missing skills and keywords.
- Help users track job applications efficiently.
- Provide actionable recommendations for career growth.

Secondary Goals

- Generate personalized interview preparation content.
 - Provide resume tailoring suggestions.
 - Offer skill gap analysis.
 - Build a portfolio-worthy AI application with production-ready architecture.
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4. Target Users

Primary Users

Final-Year Students

Need guidance on creating industry-ready resumes and preparing for placements.

Fresh Graduates

Need help optimizing resumes and tracking applications.

Entry-Level Developers

Need ATS optimization and job matching assistance.

Career Switchers

Need skill gap analysis and role-specific recommendations.

Secondary Users

Experienced Professionals

Need resume optimization and application management tools.

5. User Personas

Persona 1: College Student

Age: 21

Goals:

- Get internship opportunities
- Improve resume quality
- Track applications Pain Points:
- No ATS knowledge
- No interview preparation strategy

Persona 2: Software Engineer

Age: 26 Goals:

- Switch companies
- Improve resume visibility Pain Points:
- Applying to many jobs
- Difficult to track applications

Persona 3: Career Switcher

Age: 30 Goals:

- Transition into tech Pain Points:
- Unsure which skills are missing
- Unsure how to tailor resume

6. Scope

MVP Scope (Version 1) The

MVP must include:

1. Authentication System
2. Resume Upload and Parsing

3. ATS Resume Analysis
4. Job Description Matching
5. Job Application Tracker
6. Dashboard Analytics **Out of Scope for MVP**
 - Interview Question Generator • Resume Tailoring Generator
 - Company Research Assistant
 - Chrome Extension
 - Mobile Application
 - RAG-Based Features

These features will be implemented in future versions.

7. Functional Requirements

Module 1: Authentication

Description

Allow users to securely create accounts and access the platform.

Features User

Registration Fields:

- Full Name
- Email
- Password
- Confirm Password

Validation:

Email must be unique

Password minimum 8 characters

Password must be hashed before storage

User Login Fields:

- Email
- Password
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Behavior:

- Generate JWT access token
- Store authentication state **User Logout** Behavior:
- Remove authentication token
- Redirect to login page

Protected Routes

Only authenticated users can access:

- Dashboard
- Resume Module
- ATS Analysis
- JD Analysis
- Job Tracker

Module 2: Resume Management

Description

Allow users to upload resumes and extract structured information. **Supported**

Formats

- PDF
- DOCX

Upload Constraints Maximum

File Size:

- 10 MB

Resume Upload Flow

User uploads file

↓

Backend extracts text

↓

AI extracts structured information



Store parsed data in database



Display analysis results

Extracted Fields

Personal Information:

- Name
- Email
- Phone
- LinkedIn
- GitHub

Professional Information:

- Skills
- Education
- Experience
- Projects
- Certifications

Resume Storage

Store:

Original file

Extracted text

Parsed JSON data

Module 3: ATS Analysis

Description

Analyze resume quality and ATS compatibility. **Inputs**

- Parsed Resume

Outputs

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ATS Score Range:

0–100

Keyword Analysis

Identify:

- Present keywords
- Missing keywords **Resume Quality Checks** Evaluate:
- Contact Information
- Skills Section
- Experience Section
- Education Section
- Project Section **Suggestions** Examples:
- Add measurable achievements
- Include missing technical skills
- Improve project descriptions **ATS Analysis Response** Return:

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- - ATS Score
 - Strengths
 - Weaknesses
 - Recommendations

Module 4: Job Description Analyzer

Description

Compare resume against a target job description. **Inputs**

- Resume
- Job Description

Job Description Input Methods

- Paste Text
- Upload File (Future)

Analysis Outputs

Match Score Range:

0–100 **Missing**

Skills

Examples:

- Docker
- AWS
- Kubernetes **Missing Keywords** Examples:

CI/CD

Microservices

Agile

Recommendations Examples:

- Add Docker project experience
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- Highlight cloud technologies

Analysis Workflow Resume

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Job Description



AI Comparison



Match Score



Recommendations

Module 5: Job Application Tracker

Description

Allow users to manage job applications.

Create Application Fields:

- Company Name
- Job Title
- Source
- Application Date
- Notes

Application Statuses

- Saved
- Applied

Interview

Offer

Rejected **User**

Actions

- Create Application

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- - Edit Application
 - Delete Application
 - Change Status
 - Add Notes

Filters

- By Status
- By Company
- By Date

Module 6: Dashboard

Description

Provide a centralized overview of user activity.

Dashboard Widgets Application

Statistics Display:

- Total Applications
- Applied
- Interviews
- Offers
- Rejections

Resume Statistics Display:

ATS Score

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- - Resume Upload Date
 - Skills Count

Recent Activity Display:

- Resume Uploads
- JD Analyses
- Application Updates **Dashboard Charts**
- Application Status Distribution
- Monthly Applications

8. User Stories

Authentication

As a user, I want to register so that I can access the platform.

As a user, I want to log in securely so that my data remains private.

Resume Management

As a user, I want to upload my resume so that I can receive ATS feedback.

ATS Analysis

As a user, I want an ATS score so that I know how recruiter-friendly my resume is.

JD Matching

As a user, I want to compare my resume against a job description so that I can improve my chances of getting shortlisted.

Job Tracker

As a user, I want to track applications so that I can manage my job search efficiently.

9. Success Metrics Technical Metrics

- Resume upload success rate > 95%
- ATS analysis completion rate > 95%
- JD analysis completion rate > 95%

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- API response time < 2 seconds (excluding AI processing)
- Authentication success rate > 99%

Product Metrics

- Users upload at least one resume
- Users perform ATS analysis
- Users perform JD matching
- Users create job applications
- Users return to dashboard multiple times

10. Non-Functional Requirements Performance

- API response time under 2 seconds for standard requests
- Dashboard loads within 3 seconds **Security**
- JWT Authentication
- Password Hashing
- Input Validation
- File Upload Validation
- Protected API Routes **Scalability**
- Modular architecture
- Service-based backend structure
- Database indexing **Reliability**
- Error handling for all APIs

Logging system

Graceful failure handling

Maintainability

- Clean folder structure
- Type-safe frontend
- Modular backend services

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- - Environment-based configuration **Deployment**
 - Dockerized Backend
 - Dockerized Frontend
 - PostgreSQL Database
 - Environment Variables
 - Production Deployment Ready
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11. Future Features (Version 2)

AI Resume Tailoring

Generate customized resume suggestions for specific jobs.

Interview Preparation

Generate personalized interview questions based on resume and job description.

Skill Gap Analyzer

Identify missing skills for target roles.

Company Research Assistant

Provide company insights before interviews.

Chrome Extension

Save jobs directly from LinkedIn and job portals.

12. Acceptance Criteria

The MVP is considered complete when:

Users can register and log in.

Users can upload PDF and DOCX resumes.

Resume parsing works successfully.

ATS score is generated.

- JD matching score is generated.
 - Missing skills are identified.
 - Users can create and manage job applications.
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- - Dashboard displays analytics.
 - Application is fully dockerized.
 - Application can be deployed to production.
 - All core APIs are documented and functional.

2. Feature List

MVP (Version 1)

Authentication

- Register
- Login
- JWT Authentication
- Protected Routes

Resume Module

- Upload PDF
- Upload DOCX
- Resume Parsing
- Resume Management

ATS Module

- ATS Score
- Keyword Detection

Resume Feedback

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JD Analyzer

- JD Upload/Paste
- Match Percentage
- Missing Skills **Job Tracker**
- Create Application
- Update Status
- Delete Application
- Notes

Dashboard

- Metrics
- Charts
- Recent Activity

Version 2

- AI Resume Tailoring
- Interview Question Generator
- Skill Gap Analyzer
- Company Research
- Chrome Extension

Changes Needed in the PRD

1. Add Application Source Analytics

Currently you store the application source, but the dashboard does not explicitly use it.

Add:

Dashboard Analytics

Application Sources

Display:

- LinkedIn Applications
- Naukri Applications
- Indeed Applications
- Company Website Applications

- Referral Applications

Metrics:

- Response Rate per Source
- Interview Rate per Source
- Offer Rate per Source

This becomes one of the most unique features in your project.

2. Add Resume Versioning

Currently:

Resume Upload

Instead:

Resume Upload & Versioning

Users can upload multiple resume versions.

Examples:

Resume_v1.pdf

Resume_v2.pdf

Backend_Resume.pdf

Users can compare ATS scores across versions.

This is much closer to a real product.

3. Add Audit Trail

In Activity Logs:

Track:

- Resume Uploaded
- ATS Analysis Generated
- JD Analysis Generated
- Application Created
- Application Status Updated

Useful for dashboard activity feeds.

4. Change Authentication Storage

Current:

Store authentication state

In SDD use:

JWT Access Token

+

Refresh Token

Store access token securely.
Use HTTP-only cookies in production.